

Algebraic Topology

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A graduate course in algebraic topology from the University of Chicago.

Definition 0.1 (Pointed Topological Space)

A **pointed topological space** is a topological space X with a distinguished point $x_0 \in X$, denoted (X, x_0) .

1 Homotopy

Definition 1.1 (Homotopy)

Let $f, g : X \rightarrow Y$ be continuous maps. f is said to be **homotopic** to g , denoted $f \simeq g$, if there exists a continuous map $H : X \times [0, 1] \rightarrow Y$ —called a *homotopy*—satisfying

$$H(x, 0) = f(x) \quad (1)$$

$$H(x, 1) = g(x) \quad (2)$$

We claim that homotopy is an equivalence relation.

1. f and G are said to be **homotopic relative to a subset** $S \subset X$ if H also satisfies

$$H(s, t) = f(s) = g(s) \quad (3)$$

for all $s \in S$.

2. f is said to be **null-homotopic** if it is homotopic to a constant function.



Figure 1: Think of the blobs as images of a certain function residing in Y . They are changing continuously.

Proof. This is clearly an equivalence relation defined on $C(X, Y)$, the set of all continuous functions from X to Y .

1. *Reflexive.* Clearly, f is homotopic to itself by setting $H(x, t) = f(x)$.
2. *Symmetric.* Given homotopy $H(x, t)$ from f to g , the homotopy $\bar{H}(x, t) := H(x, 1 - t)$ maps from g to f .
3. *Transitive.* Given homotopy H_1 from f to g , and homotopy H_2 from g to h , the homotopy defined

$$H(x, t) := \begin{cases} H_1(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ H_2(x, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases} \quad (4)$$

is indeed a homotopy from F_1 to F_3 . To prove continuity, note that $H_1(x, 2t)$ is continuous over the closed set $X \times [0, \frac{1}{2}]$ and $H_2(x, 2t - 1)$ is continuous over the closed set $X \times [\frac{1}{2}, 1]$, so by the gluing lemma H is continuous.

Definition 1.2 (Homotopy Class)

The space of homotopy classes from topological space X to Y is denoted

$$[X, Y] := \frac{C^0(X, Y)}{\sim} \quad (5)$$

where $\sim$ is the homotopy relation.

Example 1.1 (Simple Homotopy)

If $f, g : \mathbb{R} \rightarrow \mathbb{R}^2$ is defined as a

$$f(x) \equiv (x, x^3), \quad g(x) \equiv (x, e^x) \quad (6)$$

then the map

$$H : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}^2, \quad H(x, t) \equiv (x, (1-t)x^3 + te^x) \quad (7)$$

is a homotopy between them.

Example 1.2 (Another Homotopy)

Let $id_B : B^n \rightarrow B^n$ be the identity function on the unit n -disk, and let $c_0 : B^n \rightarrow B^n$ be the 0-function sending every vector to 0. Then, id_B and c_0 are homotopic, with homotopy explicitly defined

$$H : B^n \times [0, 1] \rightarrow B^n, \quad H(x, t) \equiv (1-t)x \quad (8)$$

The first realization is that path homotopies are trivial if the codomain is convex.

Example 1.3 (Functions Mapping to Convex Spaces are Null-Homotopic and are Homotopic)

If $Y = \mathbb{R}^n$, then for any space X , any two continuous maps $f_0, f_1 : X \rightarrow \mathbb{R}^n$ are homotopic by defining the *linear path homotopy*

$$F(x, t) = (1-t)f_0(x) + tf_1(x) \quad (9)$$

which is well-defined as Y is convex. Furthermore, it is path connected, and so the value of the constant function does not matter. Any two constant functions are homotopic.^a

^aLet $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be 2 continuous functions. Then $f \simeq g$, since we can construct $F : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ defined $F(x, t) \equiv (1-t)f(x) + tg(x)$. (Note that the set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ is a convex set.)

Example 1.4 (Two Non-Homotopic Functions)

Consider $X = S^1, Y = \mathbb{R}^2 \setminus \{0\}$, with $f : S^1 \rightarrow Y$ the canonical inclusion, and g any constant map. Then the straight line homotopy no longer works, since the circle gets “stuck” on the hole. That gives some sort of rigorous definition of what a hole can be.

However, note that if two functions are null-homotopic, it *does not* mean that they are homotopic to each other.

Example 1.5 (Null-Homotopic Paths are Not Homotopic)

Given $X = \mathbb{R}, Y = (0, 1) \cup (2, 3)$, say that we are given the functions $f, g : X \rightarrow Y$ defined $f(x) = 0.5, f(y) = 2.5$. Then by definition they are null-homotopic, but they are not homotopic to each other.

Example 1.6 (Null-Homotopic Function)

Take a look at a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, which represents an arbitrary surface in $\mathbb{R}^2 \oplus \mathbb{R}$. Now, observe the constant function $c(x, y) \equiv c$, which represents a plane parallel to the x, y -plane. Clearly, we can

imagine a deformation of the surface of f to the flat surface of c with the homotopy

$$H(x, t) \equiv t f(x) + (1 - t)c(t) \quad (10)$$

which visually represents a linear deformation of c to f . Therefore, f is null-homotopic.

Lemma 1.1 (Continuous Compositions of Homotopic Maps are Homotopic)

Let $f_0, f_1 : X \rightarrow Y$ be continuous maps with $f_0 \simeq_A f_1$.

1. If $g : Y \rightarrow Z$ is continuous, then $g \circ f_0 \simeq_A g \circ f_1$.
2. If $h : W \rightarrow X$ is continuous, then $f_0 \circ h \simeq_{h^{-1}(A)} f_1 \circ h$.

Proof. $f_0 \simeq f_1$ implies that there is a homotopy

$$F : X \times [0, 1] \rightarrow Y \quad (11)$$

s.t. $F(x, 0) = f_0(x), F(x, 1) = f_1(x)$.

1. Now we construct the map

$$G = g \circ F : X \times [0, 1] \rightarrow Z \quad (12)$$

Then, g continuous implies G is continuous, and we have

$$G(x, 0) = (g \circ F)(x, 0) = g(f_0(x)) = (g \circ f_0)(x) \quad (13)$$

$$G(x, 1) = (g \circ F)(x, 1) = g(f_1(x)) = (g \circ f_1)(x) \quad (14)$$

which is by definition a homotopy.

2. Given $h : W \rightarrow X$ continuous, then $h \times \text{id} : W \times [0, 1] \rightarrow X \times [0, 1]$ is continuous under the product topology, and so we define the continuous map

$$H = F \circ (h \times \text{id}) : W \times [0, 1] \rightarrow YH \quad (15)$$

which satisfies

$$H(x, 0) = (F \circ (h \times \text{id}))(x, 0) = F(h(x), 0) = f_0(h(x)) = (f_0 \circ h)(x) \quad (16)$$

$$H(x, 1) = (F \circ (h \times \text{id}))(x, 1) = F(h(x), 1) = f_1(h(x)) = (f_1 \circ h)(x) \quad (17)$$

Now we have seen that homotopies form an equivalence class. Now it turns out that under function compositions, homotopies are preserved, making homotopies a congruence class.

Theorem 1.2 (Homotopies is a Congruence Relation Under Composition)

Given $f_0, f_1 : X \rightarrow Y$ and $g_0, g_1 : Y \rightarrow Z$,

$$f_0 \simeq f_1, g_0 \simeq g_1 \implies g_0 \circ f_0 \simeq g_1 \circ f_1 \quad (18)$$

Proof. We have $F : [0, 1] \times X \rightarrow Y$ and $G : [0, 1] \times Y \rightarrow Z$. Then, we can simply define the component-wise composition of maps

$$G \circ (F \times \text{id}) : [0, 1] \times X \rightarrow Z, \quad (G \circ (F \times \text{id}))(t, x) := G(t, F(t, x)) \quad (19)$$

which is continuous. Also, it is the case that

$$(G \circ F)(0, x) = G(0, F(0, x)) = G(0, f_0(x)) = g_0(f_0(x)) = (g_0 \circ f_0)(x) \quad (20)$$

$$(G \circ F)(1, x) = G(1, F(1, x)) = G(1, f_1(x)) = g_1(f_1(x)) = (g_1 \circ f_1)(x) \quad (21)$$

So this is indeed a homotopy between $g_0 \circ f_0 \simeq g_1 \circ f_1$.

1.1 Path Homotopies and the Fundamental Group

Even though we have defined what a path is, let's redefine it with some other useful properties.

Definition 1.3 (Path)

A **path** is a continuous^a map $f : [0, 1] \rightarrow X$.

1. A **reparameterization** of a path is defined by a continuous map $\varphi : [0, 1] \rightarrow [0, 1]$ satisfying $\varphi(0) = 0, \varphi(1) = 1$. $f \circ \varphi$ is known as a *reparameterized path*.
2. The **reverse path** of f is defined as $\bar{f}(t) = f(1 - t)$.
3. A **loop** is a path f with $f(0) = f(1) = x_0$, called a *base point*.

^aWe will assume the usual Euclidean subspace topology on $[0, 1]$.

Proof. A reparameterization is continuous since composition of continuous functions are continuous. The reverse is also continuous.

This leads to our definition of *path homotopies*, which is just a specific type of homotopy.

Definition 1.4 (Path Homotopy)

Given two paths $f_0, f_1 : [0, 1] \rightarrow X$, a **path homotopy** is a homotopy relative to $\{0, 1\}$, i.e. a continuous map $F : [0, 1] \times [0, 1] \rightarrow X$ s.t.

$$f(s, 0) = f_0(s) \quad f(s, 1) = f_1(s) \quad (22)$$

$$f(0, t) = x \quad f(1, t) = x \quad (23)$$

for all s, t .^a

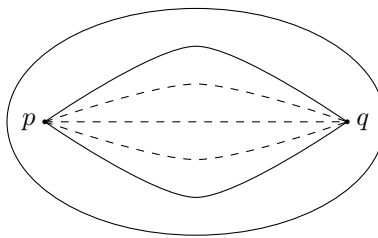


Figure 2: We can visualize two paths (sharing the same endpoints) being path homotopic if we can “continuously deform” one onto another.

^aNote that the first line is the regular homotopy conditions, while the second line is the “relative to”.

Corollary 1.3 (Reparameterizations are Still Path Homotopic)

Suppose that $g : [0, 1] \rightarrow [0, 1]$ is a continuous map with $g(0) = 0, g(1) = 1$. Let $f : [0, 1] \rightarrow X$ be a path, and $f' = f \circ g$. Then $f \simeq_p f'$.

Proof. We can take the homotopy

$$G : [0, 1] \times [0, 1] \rightarrow [0, 1], \quad G(s, t) := (1 - t)g(s) + ts \quad (24)$$

This is a homotopy between g and the identity map $[0, 1]$ relative the endpoints since $G(s, 0) = g(s)$ and $G(s, 1) = s$. Now we compose this with f to get the new continuous map

$$F = f \circ G : [0, 1] \times [0, 1] \rightarrow X, \quad (25)$$

where

$$F(s, 0) = f(G(s, 0)) = f(g(s)) = (f \circ g)(s) \quad (26)$$

$$F(s, 1) = f(G(s, 1)) = f(s) = f(s) \quad (27)$$

and so $f \circ g \simeq_p f$.

The goal here is to try and develop some sort of algebraic structure on these paths. Therefore, let us define the following binary operation.

Definition 1.5 (Composition of Paths)

Given two paths $f, g : [0, 1] \rightarrow X$ with $f(1) = g(0)$, the composition of paths is defined

$$f * g : [0, 1] \rightarrow X, \quad (f * g)(s) := \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(2s - 1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \quad (28)$$

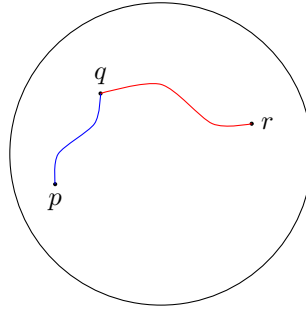


Figure 3: It is easy to visualize the product of two paths as the longer path created by "connecting" the two smaller paths.

Proof. $f * g$ is a continuous by the gluing lemma.

Note that to compose paths f and g , we must have $f(1) = g(0)$. Therefore, path composition is a partially defined operator on the set of paths, which is a bit ugly. If we focus on the set of all *loops* around x_0 , then we can guarantee that their composition is still a loop around x_0 . Moreover, we can upgrade this to a *congruence relation*.

Definition 1.6 (Homotopy Congruence on Loops)

Let (X, x_0) be a pointed topological space, with loops f, g about x_0 . Then, we define the composition operation on the homotopy equivalence classes of f and g as

$$[f] * [g] := [f * g] \quad (29)$$

This turns $\simeq$ into a congruence relation.

Proof. We must show that this operation is well-defined. Let $f_0 \sim f_1$ and $g_0 \sim g_1$. Then, by definition, there exists homotopies

$$F(s, t), \quad F(s, 0) = f_0, F(s, 1) = f_1 \quad (30)$$

$$G(s, t), \quad G(s, 0) = g_0, G(s, 1) = g_1 \quad (31)$$

$$(32)$$

We wish to connect the two functions $f_0 * g_0$ and $f_1 * g_1$ with a homotopy. These are just two loops anyways, so if we compose the homotopies as

$$H(s, t) := (F_t * G_t)(s) \quad (33)$$

which is a homotopy since it is continuous, with $H(s, 0) = (F_0 * G_0)(s) = (f_0 * g_0)(s)$, $H(s, 1) = (F_1 * G_1)(s) = (f_1 * g_1)(s)$, and as for the endpoints, $H(0, t) = H(1, t) = x_0$.

1.2 Fundamental Group**Definition 1.7 (Fundamental Group)**

Let (X, x_0) be a pointed topological space, and denote the set of congruence classes of all loops about x_0 as $\pi_1(X, x_0)$. This set is an group under path composition, called the **fundamental group**.

Proof. We prove the group axioms. We have just proved that path composition of equivalence classes $[f] * [g] := [f * g]$ is well-defined.

1. *Closed.* Clearly, if f, g are loops, then $f * g$ is a loop, so $[f] * [g] = [f * g]$ is a loop.
2. *Identity.* The identity element is the equivalence class of the constant map $\iota(s) = x_0$. We can see that given a path f ,

$$c * f = \begin{cases} x_0 & \text{if } 0 \leq s \leq \frac{1}{2} \\ f(2s - 1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}, \quad f * c = \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ x_0 & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \quad (34)$$

are reparamaterizations of f , so $[c * f] = [f * c] = [f]$.

3. *Inverses.* Given $[f]$, take a representative path f and take the reversed path $\bar{f}$. Then, we wish to show that $f * \bar{f}$ and $\bar{f} * f$ is homotopic to a constant path. This is a little tricky since we can't just use reparamaterizations, but let us define the homotopy from f to the constant function first.

$$f_t(s) = \begin{cases} f(s) & \text{if } 0 \leq s \leq 1 - t \\ f(1 - t) & \text{if } 1 - t \leq s \leq 1 \end{cases} \quad (35)$$

Then, we can define the product

$$H_t(s) := (f_t * \bar{f}_t)(s) \quad (36)$$

which is our desired homotopy. Therefore, $[f] * [\bar{f}] = [f * \bar{f}] = [x_0]$.

4. *Associativity.* Given $[f], [g], [h]$, we wish to show that

$$[f] * ([g] * [h]) = [f * (g * h)] = [(f * g) * h] = ([f] * [g]) * [h] \quad (37)$$

Let's take representative elements $f \in [f], g \in [g], h \in [h]$. We could do this directly by defining a homotopy from $f * (g * h)$ to $(f * g) * h$, but this becomes too messy to write.^a Rather, it suffices to show that if one is a reparameterization of the other, then we are done. We can reparameterize

$$(f * (g * h)) = ((f * g) * h) \circ \varphi, \quad \varphi(s) = \begin{cases} 2s & \text{if } 0 \leq s \leq \frac{1}{4} \\ s + \frac{1}{4} & \text{if } \frac{1}{4} \leq s \leq \frac{1}{2} \\ \frac{s}{2} + \frac{1}{2} & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \quad (38)$$

^aI think the actual homotopy is $F(s, t) = \begin{cases} f((t+1)s) & \text{if } 0 \leq s \leq \frac{1}{4} \\ g(s + \frac{t}{4}) & \text{if } \frac{1}{4} \leq s \leq \frac{1}{2} \\ h((1 - \frac{t}{2})s + \frac{t}{2}) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$.

Note that the identity element of $\pi_1(X, x_0)$ is the *constant* map $x \mapsto x_0$, not the identity map! This is different from a group of transformations. We can immediately compute the fundamental group for convex sets.

Example 1.7 (Fundamental Group of Convex Sets are Trivial)

If C is a convex set and $x \in C$, $\pi_1(C, x) = \{e\}$, the trivial group. This is true since every loop is null-homotopic to the constant point at x through a linear homotopy.

It isn't so easy to show that a space has a nontrivial fundamental group since one must somehow demonstrate the nonexistence of homotopies between certain loops. But we can gain some intuition through some visuals.

Example 1.8 (Visualization of the Fundamental Group of the Annulus)

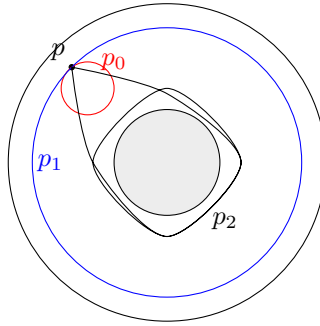


Figure 4: Consider the space $X \equiv B_2 \setminus B_1$, which is the 2-disk without the unit disk in $\mathbb{R}^2$. Given an arbitrary point $p \in X$, there exists an infinite number of path classes of X at p , denoted $[p_i]$, where i corresponds to how many times the paths loop around the hole. The first three path classes are shown below. It is clear that $[p_0]$ is the identity, and the group operation rule is $[p_i] \cdot [p_j] = [p_{i+j}]$ meaning that $\pi_1(X, p)$ is the infinite discrete group generated by $[p_0]$ and $[p_1]$.

1.3 Induced Homomorphisms

Now that we have constructed the fundamental group at a point x_0 , the next things to consider are their behavior under group homomorphisms, and how different fundamental groups at different points (of the same topological space) look like.

Definition 1.8 (Induced Group Homomorphism)

Suppose $k : X \rightarrow Y$ is a continuous map with $x_0 \in X, y_0 = f(x_0) \in Y$. Define the map, called the **induced group homomorphism** of k as

$$k_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0), \quad k_*([f]) := [k \circ f] \quad (39)$$

which is well defined since $f \simeq_p f'$ in $X \implies k \circ f \simeq_p k \circ f'$ in Y as a composition with continuous functions. This is indeed a group homomorphism.

Proof. Let's take a look at $f, g : [0, 1] \rightarrow X$. Then

$$f * g : [0, 1] \rightarrow X, \quad (f * g)(t) := \begin{cases} f(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases} \quad (40)$$

Therefore, mapping this through k_* gives us

$$k \circ (f * g) : [0, 1] \rightarrow Y, \quad (k \circ (f * g))(t) := \begin{cases} (k \circ f)(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ (k \circ g)(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases} \quad (41)$$

Now take $k \circ f, k \circ g : [0, 1] \rightarrow Y$, and we can construct

$$(k \circ f) * (k \circ g) : [0, 1] \rightarrow Y, \quad ((k \circ f) * (k \circ g))(t) = \begin{cases} (k \circ f)(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ (k \circ g)(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases} \quad (42)$$

which matches. Therefore we have $k \circ (f * g) = (k \circ f) * (k \circ g)$, and so under the congruence class

$$k_*([f] * [g]) = k_*([f * g]) = [k \circ f] * [k \circ g] = k_*([f]) * k_*([g]) \quad (43)$$

Lemma 1.4 (Identity Map Induces Identity Group Homomorphism)

If the continuous map between topological spaces is $k = \text{id}_X$, then the induced homomorphism is simply the identity isomorphism.

$$(\text{id}_X)_* = (\text{id}_{\pi_1(X, x_0)}) \quad (44)$$

Proof. Just substituting with the formula above gives

$$k_*([f] * [g]) = [f] * [g] \quad (45)$$

and so k_* must be the identity.

Theorem 1.5 (Composition of Induced Group Homomorphisms)

If $k : X \rightarrow Y$ and $l : Y \rightarrow Z$ are continuous, then

$$(l \circ k)_* = l_* \circ k_* \quad (46)$$

Proof. We have

$$(l \circ k)_*([f]) = [(l \circ k) \circ f] \quad (47)$$

$$= [l \circ (k \circ f)] \quad (48)$$

$$= l_*([k \circ f]) \quad (49)$$

$$= l_*(k_*([f])) \quad (50)$$

$$= (l_* \circ k_*)([f]) \quad (51)$$

Corollary 1.6 (Homeomorphisms Induce Fundamental Group Isomorphisms)

If $X \cong Y$ by a homeomorphism taking x_0 to y_0 , then $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$.

Proof. If $k : X \rightarrow Y$ is a homeomorphism, then $k : Y \rightarrow X$ is also a homeomorphism, both continuous. Then this induces the group homomorphisms $k_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ and $k_*^{-1} : \pi_1(Y, y_0) \rightarrow \pi_1(X, x_0)$, and so from the lemma,

$$k^{-1} \circ k = \text{id}_X \implies (k^{-1} \circ k)_* = \text{id}_{\pi_1(X, x_0)} \quad (52)$$

and so $(k^{-1})_* \circ k_* = \text{id}_{\pi_1(X, x_0)}$ and $k_* \circ (k^{-1})_* = \text{id}_{\pi_1(Y, y_0)}$.

Finally, let's talk about constructing new groups from old ones.

Theorem 1.7 (Direct Products of Fundamental Groups)

For any spaces X, Y , we have

$$\pi_1(X \times Y, x_0 \times y_0) \simeq \pi_1(X, x_0) \times \pi_1(Y, y_0) \quad (53)$$

Moreover, the inclusion and projection maps on the topological spaces induce inclusion and projection maps on the groups.

Proof. A loop $f : I \rightarrow X \times Y$ with $f(0) = f(1) = (x_0, y_0)$ is the same as a pair of maps $f_X : I \rightarrow X$ and $f_Y : I \rightarrow Y$. A homotopy between loops is a continuous map $F : I \times I \rightarrow X \times Y$, which is the same as a pair of homotopies F_X and F_Y .

A fancy way to say this is that at every point, there is a group, and π_1 is a functor from the category of pointed topological spaces to the category of groups.

1.4 Simply Connected Spaces

Now the final thing to mention for now is that there is a group at every single point in a topological space. They may be completely different groups, but the following theorem nicely categorizes them in terms of path components.

Theorem 1.8 (Fundamental Groups of Points in a Path Component are Isomorphic)

If x_0 and x_1 are on the same path component of X , then $\pi_1(X, x_0) \simeq \pi_1(X, x_1)$. The actual isomorphism depends on a choice of a path from x_0 to x_1 .

Proof. Let us construct a path $\gamma : [0, 1] \rightarrow X$ s.t. $\gamma(0) = x_0, \gamma(1) = x_1$. Then, let $[f] \in \pi_1(X, x_0)$ be a loop about x_0 . We can see that $\gamma * f * \bar{\gamma} \pi_1(X, x_1)$ must be a loop around x_1 . Similarly, if $[f] \in \pi_1(X, x_1)$, then $\bar{\gamma} * f * \gamma \in \pi_1(X, x_0)$. This establishes a bijection between $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$. It suffices to show that the map $A : [f] \mapsto [\gamma * f * \bar{\gamma}]$ is a group homomorphism.

Essentially, we want to start at x_1 , go to x_0 , loop around, and then go back to x_1 , which is in total a loop around x_1 . We claim that A_h is an isomorphism by showing that it is a group homomorphism.

$$A_h([f] * [g]) = [\bar{h} * f * g * h] \quad (54)$$

$$= [\bar{h} * f] * [g * h] \quad (55)$$

$$= [\bar{h} * f] * [h * \bar{h}] * [g * h] \quad (56)$$

$$= [\bar{h} * f * h] * [\bar{h} * g * h] \quad (57)$$

$$= A_h([f]) * A_h([g]) \quad (58)$$

Of course, when x_0 and x_1 are in different path components, they may or may not have isomorphic fundamental groups. Therefore, when we know a space X is path connected, we can just drop the point x_0 and just refer to the fundamental group as $\pi_1(X)$.

Therefore, if the space is path connected, the fundamental group at all points are isomorphic. But even if all the groups are isomorphic, they may be nontrivial. This motivates the following definition.

Definition 1.9 (Simply Connected)

A path connected topological space X is **simply connected** if it meets each of the equivalent requirements.

1. It has a trivial fundamental group.^a
2. For any two points $p, q \in X$, there is a unique homotopy class of paths connecting p to q .

^aDoesn't matter the base point, since it is path connected already.

Proof. We prove bidirectionally.

1. ($\rightarrow$). Let $\gamma, \phi : [0, 1] \rightarrow X$ be two paths from p to q . Then, $\gamma * \bar{\phi}$ must be null-homotopic since the fundamental group is trivial. Therefore,

$$[\gamma] * [\bar{\phi}] = [\gamma * \bar{\phi}] = [e] = [\gamma * \bar{\gamma}] = [\gamma] * [\bar{\gamma}] \quad (59)$$

This implies that $[\bar{\phi}] = [\bar{\gamma}] \implies \bar{\phi} \simeq \bar{\gamma} \implies \phi \simeq \gamma \implies [\phi] = [\gamma]$.

2. ($\leftarrow$). Let f be a loop around p . We wish to show that f is null-homotopic. If f is constant, we are done. If not, then there exists a $q \neq p$ where $q \in \text{Im}(f)$. We can then reparameterize f into components

$$f = f_1 * f_2, \quad \begin{cases} f_1(0) = p, f_1(1) = q \\ f_2(0) = q, f_2(1) = p \end{cases} \quad (60)$$

Since there is a unique homotopy class, it must be the case that $[\bar{f}_1] = [f_2]$. Therefore,

$$[f] = [f_1 * f_2] = [f_1] * [f_2] = [f_1] * [\bar{f}_1] = [e] \quad (61)$$

Note that the actual isomorphism depends on the choice of the path γ . Note the following.

$$\text{Simply Connected} \implies \text{Path Connected} \implies \text{Connected} \quad (62)$$

Example 1.9 (Simply Connected Spaces)

The following are simply connected.

1. $\mathbb{R}^n$
2. Any convex subset of $\mathbb{R}^n$.
3. Any star-shaped subset of $\mathbb{R}^n$, where a star-shaped set is defined as a set where there exists some point $x_0 \in A$ s.t. for all $y \in A$ the line segment from x_0 to y is in A .

Theorem 1.9 (Union of Simply Connected Spaces)

Suppose $X = U \cup V$, where U, V are open and simply connected and $U \cap V$ is nonempty and path connected. Then X is simply connected.

Proof. Let $x_0 \in U \cap V$. Given a loop f , we can factor f as a composition $f = f_1 * \cdots * f_n$, where each f_i is a path contained either in U or in V , from some $x_{i-1} \in U \cap V$ to $x_i \in U \cap V$, and $x_n = x_0$. For $i = 1, \dots, n-1$, choose a path g_i from x_0 to x_i in $U \cap V$.

$$f \simeq_p f_1 \bar{g}_1 g_1 f_2 \bar{g}_2 g_2 f_3 \cdots f_{n-1} \bar{g}_{n-1} g_{n-1} f_n \quad (63)$$

$$\simeq_p (f_1 \bar{g}_1)(g_1 f_2 \bar{g}_2)(g_2 f_3 \cdots f_{n-1} \bar{g}_{n-1})(g_{n-1} f_n) \quad (64)$$

Each of the factors is a loop in U or V based at x_0 , so it can be contracted to e_{x_0} . Thus, $f \simeq e_{x_0}$, as required.

1.5 Covering Maps

Now it turns out that a lot of sets are simply connected. So we will use this definition to categorize the fundamental groups of familiar spaces. However, the proofs of these requires the knowledge of covering spaces and lifts.

Definition 1.10 (Covering Map)

A function $p : \tilde{X} \rightarrow X$ is a **covering map** if p is

1. surjective,
2. continuous,
3. and for every $x \in X$ there exists an open $U \ni x$ s.t. $p^{-1}(U) = \cup_{\alpha} V_{\alpha}$, where V_{α} are open and disjoint, and $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$ is a homeomorphism.

The sets $\{V_{\alpha}\}$ is a partition into **slices**, and $(\tilde{X}, p)$ is said to be a **covering space** of X , and X is said to be **evenly covered**.

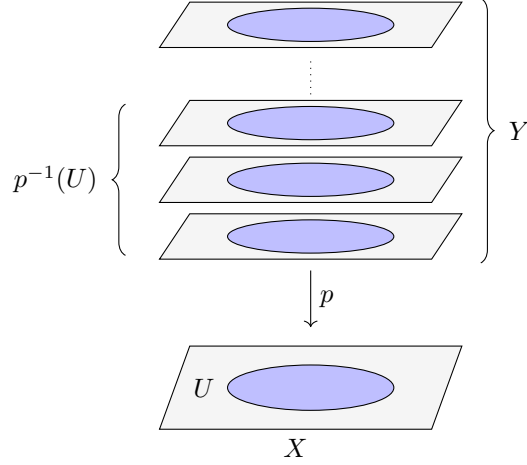


Figure 5: The best way to visualize this is to think of a “stack of pancakes.” Consider $Y = X \times \mathbb{Z}$ and let $p(x, z) = x$ be a projection. Given some open $U \subset X$, $p^{-1}(U) = U \times \mathbb{Z}$, with each slice $V_z = U \times \{z\}$ mapping homeomorphically onto U .

Example 1.10 (Trivial Covering Map)

The most trivial covering map is when $\tilde{X} = X$ and $p = \text{id}_X$.

Example 1.11 (Covering Map of S^1 to Itself)

We can consider $S^1 \subset \mathbb{C}$ and claim that the following is a covering map.

$$p : S^1 \rightarrow S^1, \quad p(z) = z^2 \quad (65)$$

This is clearly continuous. It is surjective since $p(e^{xi}) = e^{2xi}$, so $p^{-1}(e^{xi}) = e^{xi/2}$. For every $x = e^{i\theta} \in X$, we wish to show that there is an open neighborhood U such that $p^{-1}(U)$ is a disjoint union of open sets V_α that each map homeomorphically to U . Well, we can take some $0 < \epsilon < \pi$ and consider the set

$$U := \{e^{it} \in X : t \in (\theta - \epsilon, \theta + \epsilon)\} \quad (66)$$

Since this arc has angle no larger than 2π , every point of U has a unique angle. Now to get $p^{-1}(U)$, we solve

$$z^2 = e^{it} \implies z = e^{it/2}, e^{i(\frac{t}{2} + \pi)} \quad (67)$$

Therefore,

$$p^{-1}(U) = \underbrace{\left\{ e^{is} : \frac{\theta - \epsilon}{2} < s < \frac{\theta + \epsilon}{2} \right\}}_{V_0} \sqcup \underbrace{\left\{ e^{is} : \frac{\theta - \epsilon}{2} + \pi < s < \frac{\theta + \epsilon}{2} + \pi \right\}}_{V_1} \quad (68)$$

These are two disjoint open arcs of S^1 . The restrictions $p|_{V_0}$ and $p|_{V_1}$ are homeomorphisms. Geometrically, S^1 upstairs wraps twice around S^1 downstairs. A sufficiently small arc downstairs therefore has exactly two separate copies upstairs.

Example 1.12 (Reals to the Circle)

Given S^1 , let us consider the covering map $p : \mathbb{R} \rightarrow S^1$ defined $p(t) = (\cos(t), \sin(t))$. It is clearly surjective and continuous. Now given any $x \in S^1$, let's take the open neighborhood $U \ni x$. Then, its preimage is the union of disjoint intervals V_n for $n \in \mathbb{N}$, and if we restrict it to each V_n , then it is a homeomorphism.

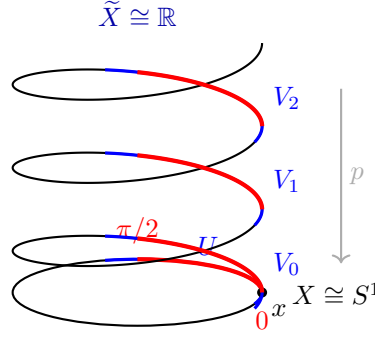


Figure 6: The covering map $p : \tilde{X} \rightarrow X$ visualized as a helix. The open arc $U \subset X$ has preimage $p^{-1}(U) = V_0 \sqcup V_1 \sqcup V_2 \sqcup \cdots$, where each V_α maps homeomorphically onto U . The red arcs indicate the portion corresponding to angles from 0 to $\pi/2$.

Example 1.13 (Positive Reals to the Circle)

The map

$$p : \mathbb{R}^+ \rightarrow S^1, \quad p(x) := (\cos 2\pi x, \sin 2\pi x) \quad (69)$$

is not a covering map. It is surjective and continuous but the point $x_0 = (1, 0) \in S^1$ has no neighborhood U that is evenly covered by p . We can see that the inverse image of $(0, \epsilon)$ is not homeomorphic to an open U .

Now let's introduce a few properties of covering maps.

Lemma 1.10 (Preimage of Singletons Have Discrete Topology)

If $p : \tilde{X} \rightarrow X$ is a covering map, then for each $x \in X$, the subspace $p^{-1}(x) \subset \tilde{X}$ has the discrete topology.

Proof. Take some $x \in X$ and look at its preimage $p^{-1}(x)$. Then, we can imagine it as some disjoint union of discrete points, which has the discrete topology. More formally, take some open $U \ni x$ and then its preimage will be the disjoint union of open sets $\sqcup_\alpha V_\alpha$. But each point $y \in p^{-1}(x)$ will be contained in exactly one of the U_α 's, and so $y = y \cap U_\alpha$ is open in itself, meaning that all the $\{y_\alpha\}$ will have the discrete topology.

Theorem 1.11 (Covering Maps are Open Maps)

If $p : \tilde{X} \rightarrow X$ is a covering map, then it is open, i.e. maps open sets to open sets.

Proof. Suppose $A \subset \tilde{X}$ is open. We wish to show that $p(A)$ is open. Choose an arbitrary $x \in p(A)$. By definition of covering maps, there must exist an open $U \ni x$ (but U not necessarily in $p(A)$) s.t. $p^{-1}(U) = \sqcup_{\alpha} V_{\alpha}$ for V_{α} open. There is a point $y \in U$ s.t. $p(y) = x$. Let U_{β} be the open slice containing y . Then, since A is open by assumption, the set $V_{\beta} \cap A$ is open in $\tilde{X}$ and hence open in V_{β} . Since p maps V_{β} homeomorphically onto U , it maps open sets to open sets and so $p(V_{\beta} \cap A)$ is open in U and hence open in X . Furthermore, it is a neighborhood of x contained in $p(A)$. Since for every $x \in p(A)$ has an open neighborhood contained in $p(A)$, $p(A)$ is thus open.

Corollary 1.12 (Covering Maps are Quotient Maps)

A covering map $p : \tilde{X} \rightarrow X$ is a quotient map.

Proof. This is immediate because open maps are quotient maps.^a

^aHowever, open doesn't mean closed!

Theorem 1.13 (Restriction of Covering Maps)

Let $p : \tilde{X} \rightarrow X$ be a covering map. If $\tilde{X}_0 \subset \tilde{X}$ and $X_0 = p^{-1}(\tilde{X}_0)$, then the restriction $p_0 : \tilde{X}_0 \rightarrow X_0$ is a covering map.

Proof. Given $x_0 \in X_0$, let $U \ni x_0$ be an open set. Then let $\{V_{\alpha}\}$ be a partition of $p^{-1}(U)$ into slices. We know that $U \cap X_0$ is an open neighborhood of x_0 in X_0 , and taking the preimage we have

$$p^{-1}(U \cap X_0) = \sqcup_{\alpha} V_{\alpha} \cap p^{-1}(X_0) = \sqcup_{\alpha} V_{\alpha} \cap \tilde{X}_0 \quad (70)$$

The $V_{\alpha} \cap \tilde{X}_0$ are disjoint, and V_{α} open in $\tilde{X}$ implies that $V_{\alpha} \cap \tilde{X}_0$ is open in $\tilde{X}_0$. Finally, we can see that p maps V_{α} homeomorphically onto U , and so p maps $V_{\alpha} \cap \tilde{X}_0$ homeomorphically onto $p(V_{\alpha}) \cap p(\tilde{X}_0) = U \cap X_0$.

Theorem 1.14 (Product of Covering Maps)

If $p : \tilde{X} \rightarrow X$ and $p' : \tilde{X}' \rightarrow X'$ are covering maps, then

$$p \times p' : \tilde{X} \times \tilde{X}' \rightarrow X \times X' \quad (71)$$

is a covering map.

Proof. Given $x \in X, x' \in X'$, let $U \ni x, U' \ni x'$ be open neighborhoods of x and x' with $\{V_{\alpha}\}, \{V'_{\alpha}\}$ the even slices of $p^{-1}, (p')^{-1}$. Then the preimage

$$(p \times p')^{-1}(U \times U') = \bigcup_{\alpha, \beta} V_{\alpha} \times V'_{\beta} \quad (72)$$

These are disjoint open sets of $\tilde{X} \times \tilde{X}'$, each of which is mapped homeomorphically onto $U \times U'$ by $p \times p'$.

Theorem 1.15 (Composition of Covering Maps)

Let $q : X \rightarrow Y$ and $r : Y \rightarrow Z$ be covering maps. If $r^{-1}(z)$ is finite for all $z \in Z$, then $p = r \circ q : X \rightarrow Z$ is a covering map.

Proof.

What we should keep in mind is that locally, covering maps look like homeomorphisms.

Example 1.14 (Projection of Products with Discrete Topology)

Let Y have the discrete topology, and $p : X \times Y \rightarrow X$ is a projection on the first coordinate. Then we claim that p is a covering map. Note that the open sets are of the form

$$X \times \{y\} \text{ for } y \in Y \quad (73)$$

Example 1.15

Say $k \in \mathbb{Z} \setminus \{0\}$ and $f_k : S^1 \rightarrow S^1$ defined

$$f_k(\cos \theta, \sin \theta) := (\cos k\theta, \sin k\theta) \quad (74)$$

This circularly stretches out the circle by a factor of k . We claim that this is a covering map.

Theorem 1.16 (Covering Maps as Homeomorphisms)

Let $p : \tilde{X} \rightarrow X$ be a covering map with X simply connected and $\tilde{X}$ path connected. Then p is a homeomorphism.

1.6 Lifts

There is a nice relationship between the fundamental group and the covering space.

Definition 1.11 (Lift)

If $p : \tilde{X} \rightarrow X$ is a covering map and $f : Z \rightarrow X$ is continuous, then a **lift** of f is a continuous map $\tilde{f} : Z \rightarrow \tilde{X}$ satisfying $p \circ \tilde{f} = f$.^a

$$\begin{array}{ccc} & & \tilde{X} \\ & \nearrow \tilde{f} & \downarrow p \\ Z & \xrightarrow{f} & X \end{array}$$

Figure 7: Commutative diagram of a lift.

^aIt is called a lift since it “lifts” the function with codomain Z to the larger covering space $\tilde{X}$.

Note that not every f has a lift. However, it turns out that the existence of liftings when p is a covering map is an important tool for studying covering spaces and the fundamental group. The path lifting property guarantees that paths can be lifted into a covering space.

Lemma 1.17 (Path Lifting Property)

Suppose $p : \tilde{X} \rightarrow X$ is a covering map with $p(y_0) = x_0$. Given any path $f : [0, 1] \rightarrow X$ with $f(0) = x_0$, there exists a unique lift $\tilde{f} : [0, 1] \rightarrow \tilde{X}$ with $\tilde{f}(0) = y_0$.

Proof. Since p is a covering map, for each point $x \in \text{Im}(f)$ there must exist some open $U_x \ni x$ s.t. $p^{-1}(U_x) = \sqcup_{\alpha} U_{\alpha,x}$ with $V_{\alpha} \cong U_x$. Since continuous images of compact sets are compact, we must have $\text{Im}(f) \subset X$ is compact, and so, we can reduce this open cover of $\text{Im}(f)$ to a finite subcover. Let's label the points $x_0, x_1, \dots, x_n \in X$ with corresponding open sets U_i covering $\text{Im}(f)$, where $x_i = f(s_i)$ such that

$$0 = s_0 < s_1 < s_2 < \dots < s_{n-1} < s_n = 1 \quad (75)$$

We take each piece $[s_{i-1}, s_i]$. Since the corresponding image $f[s_{i-1}, s_i]$ is contained within $U_i \subset X$,^a by the covering property we must have

$$p^{-1}(U_i) = \bigsqcup_{\alpha} U_{\alpha,i} \quad (76)$$

If $i = 1$, then pick any α , otherwise pick the $V_{\alpha,i}$ that coincides with the image of the previous piece. Then, since $p|_{V_{\alpha}} : V_{\alpha,i} \rightarrow U_i$ is a homeomorphism, we can define

$$\tilde{f}_i : [s_{i-1}, s_i] \rightarrow V_{\alpha,i}, \quad \tilde{f}_i = (p|_{V_{\alpha}})^{-1} \circ f \quad (77)$$

Now since each $\tilde{f}_i$ is continuous over a closed set, by gluing lemma we can combine all of them to continuous $\tilde{f} : [0, 1] \rightarrow \tilde{X}$.

^aApparently I can't assume this since the intervals may not be small enough? Apparently I need Lebesgue covering lemma?

Therefore, this essentially means that by taking the covering map p that sends y_0 to x_0 and a path f starting at x_0 , we can construct a corresponding path $\tilde{f}$ starting at y_0 such that $f = p \circ \tilde{f}$. Let's do an example.

Example 1.16 (Lift of Real Interval to S^1)

Consider the covering map and path defined as below.

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = (\cos 2\pi t, \sin 2\pi t) \quad (78)$$

We can see that $p(y_0) = p(0) = (1, 0) = x_0$. Now let's consider a few paths $f : [0, 1] \rightarrow S^1$. We would like to "lift" this path $I \rightarrow S^1$ to a path $I \rightarrow \mathbb{R}$. The lemma states that there is indeed a unique path.

1. When $f(s) = (\cos \pi s, \sin \pi s)$ which satisfies $f(0) = x_0 = (1, 0)$, the lift is $\tilde{f}(t) = t/2$ which satisfies $\tilde{f}(0) = y_0 = 0$.

$$t \xrightarrow{\tilde{f}} \frac{t}{2} \xrightarrow{p} (\cos 2\pi \frac{t}{2}, \sin 2\pi \frac{t}{2}) = (\cos \pi t, \sin \pi t) = f(t) \quad (79)$$

2. When $f(s) = (\cos \pi s, -\sin \pi s)$ which satisfies $f(0) = x_0 = (1, 0)$, the lift is $\tilde{f}(t) = -t/2$ which satisfies $\tilde{f}(0) = y_0 = 0$.

$$t \xrightarrow{\tilde{f}} -\frac{t}{2} \xrightarrow{p} (\cos -2\pi \frac{t}{2}, \sin -2\pi \frac{t}{2}) = (\cos \pi t, \sin -\pi t) = f(t) \quad (80)$$

since $\cos(t) = \cos(-t)$.

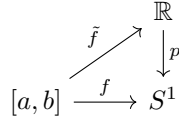


Figure 8

Then, we show that path homotopies can be lifted as well.

Lemma 1.18 (Homotopy Lifting Property)

Suppose $p : \tilde{X} \rightarrow X$ is a covering map. If paths $f, g : [0, 1] \rightarrow X$ are path homotopic relative to their endpoints, and $\tilde{f}, \tilde{g} : I \rightarrow \tilde{X}$ are lifts starting at the same y_0 , then $\tilde{f} \simeq_p \tilde{g}$.

Proof. The key is to lift the entire homotopy, not just each time-slice separately. Suppose $F : I \times I \rightarrow X$ is a homotopy from f to g relative to endpoints. We already have the lift $\tilde{f}$ of the bottom edge: $F(s, 0) = f(s)$. The idea is to try to think of $I \times I$ as a square and construct $\tilde{F} : I \times I \rightarrow \tilde{X}$ such that

$$p \circ \tilde{F} = F, \quad \tilde{F}(s, 0) = \tilde{f}(s) \quad (81)$$

For each point $(s, t) \in I \times I$, since p is a covering map, there exists an evenly covered neighborhood $U_{s,t}$ of $F(s, t)$. Hence

$$\{F^{-1}(U_{s,t})\}_{(s,t) \in I \times I} \quad (82)$$

is an open cover of $I \times I$. Since $I \times I$ is compact, by the Lebesgue number lemma there exists $\delta > 0$ such that every subset of $I \times I$ of diameter less than δ is contained in some $F^{-1}(U_{s,t})$. Choose subdivisions

$$0 = s_0 < s_1 < \cdots < s_m = 1, \quad 0 = t_0 < t_1 < \cdots < t_n = 1 \quad (83)$$

fine enough that every rectangle

$$R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j] \quad (84)$$

has diameter less than δ . Thus, for each i, j , there exists an evenly covered neighborhood $U_{ij} \subset X$ such that $F(R_{ij}) \subset U_{ij}$. Write

$$p^{-1}(U_{ij}) = \bigsqcup_{\alpha} V_{\alpha, ij}, \quad (85)$$

where each restriction $p|_{V_{\alpha, ij}} : V_{\alpha, ij} \rightarrow U_{ij}$ is a homeomorphism. We now construct $\tilde{F}$ rectangle by rectangle, beginning with the bottom row. For a rectangle R_{i1} , the bottom edge has already been lifted by $\tilde{f}$. Since $F(R_{i1}) \subset U_{i1}$, the point $\tilde{f}(s_{i-1})$ lies in exactly one sheet $V_{\alpha, i1}$ over U_{i1} . Define

$$\tilde{F}|_{R_{i1}} = (p|_{V_{\alpha, i1}})^{-1} \circ F. \quad (86)$$

This agrees with $\tilde{f}$ along the bottom edge by uniqueness of path lifting. Proceeding from left to right and then row by row, suppose $\tilde{F}$ has already been defined on the rectangles preceding R_{ij} . Choose the unique sheet $V_{\alpha, ij}$ containing the already-defined lift of a point on the shared edge of R_{ij} . Then define

$$\tilde{F}|_{R_{ij}} = (p|_{V_{\alpha, ij}})^{-1} \circ F. \quad (87)$$

On the shared edge, both this map and the previously defined map are lifts of the same path and agree at one point. By uniqueness of path lifting, they agree on the entire shared edge.

Therefore the locally defined lifts agree on all overlaps, and by the gluing lemma they combine to give a continuous map

$$\tilde{F} : I \times I \rightarrow \tilde{X} \quad (88)$$

satisfying

$$p \circ \tilde{F} = F, \quad \tilde{F}(s, 0) = \tilde{f}(s). \quad (89)$$

Now consider the top edge $h(s) = \tilde{F}(s, 1)$. Then

$$p(h(s)) = F(s, 1) = g(s), \quad (90)$$

so h is a lift of g . Since F is a homotopy relative to endpoints, the left edge satisfies $F(0, t) = x_0$ for all $t \in I$. Hence the path $t \mapsto \tilde{F}(0, t)$ is a lift of the constant path at x_0 beginning at y_0 , and therefore it is constant. In particular,

$$h(0) = \tilde{F}(0, 1) = y_0. \quad (91)$$

Thus h and $\tilde{g}$ are lifts of g with the same initial point, so by uniqueness of path lifting,

$$h = \tilde{g}. \quad (92)$$

Similarly, because F is relative to endpoints, the paths

$$t \mapsto \tilde{F}(0, t), \quad t \mapsto \tilde{F}(1, t) \quad (93)$$

are constant. Hence $\tilde{F}$ is a homotopy relative to endpoints from $\tilde{f}$ to $\tilde{g}$. Therefore $\tilde{f} \simeq_p \tilde{g}$.

Great, so we have defined covering maps and have shown that paths admit lifts that are consistent with the path homotopy relation. Now, to connect this back to fundamental groups, we introduce the following theorem which allows us to determine the cardinality of fundamental groups through covering maps over simply connected spaces. This will be very useful to us later on.

Theorem 1.19 (Bijection from Fundamental Group to Preimage of Covering Map on Simply Connected Space)

If $p : \tilde{X} \rightarrow X$ is a covering map and $\tilde{X}$ is simply connected, then $\pi_1(X, x_0)$ is in bijection with $p^{-1}(x_0)$.

Proof. Given a loop $f : [0, 1] \rightarrow X$ with $f(0) = f(1) = x_0$, fix $y_0 \in p^{-1}(x_0)$. By the path lifting property, we can take the lifted path $\tilde{f} : [0, 1] \rightarrow \tilde{X}$ with $\tilde{f}(0) = y_0$.

$$p(\tilde{f}(1)) = f(1) = y_0, \quad \tilde{f}(1) \in p^{-1}(x_0) \quad (94)$$

By the homotopy lifting property, this only depends on $[f] \in \pi_1(X, x_0)$.

1. To prove surjectivity, since $\tilde{X}$ is simply connected, this implies $\tilde{X}$ is path connected. Choose a path g from y_0 to y . Let $f = p \circ g$. Then $L([f]) = g(1) = y$.
2. To prove injectivity, if $L([f]) = L([g])$, then $\tilde{f}$ and $\tilde{g}$ are both paths in $\tilde{X}$ from y_0 to y_1 . Because $\tilde{X}$ is simply connected, $\tilde{f} \simeq_p \tilde{g} \implies f \simeq_p g \implies [f] = [g]$.

Example 1.17

Take $p : \mathbb{R}^2 \rightarrow S^1 \times S^1$ defined

$$(s, t) \mapsto ((\cos 2\pi s, \sin 2\pi s), (\cos 2\pi t, \sin 2\pi t)) \quad (95)$$

Think back to the quotient space representation.

$$T^2 = [0, 1]^2 / \sim, \quad (x, 0) \sim (x, 1) \forall x, (0, y) \sim (1, y) \forall y \quad (96)$$

Then we can think of $p(x, y) = (x \pmod{1}, y \pmod{1})$. We claim that this is a covering map.

1.7 Fundamental Groups of Euclidean Space and the Sphere

Now that we have established that $\mathbb{R}^n$ is simply connected, along with some tools on sufficient conditions for simply connectedness and lifts, we can start to categorize fundamental groups of $\mathbb{R}^n$ and S^n , with the corresponding sets minus a point.

Theorem 1.20 (Fundamental Group of S^1)

$$\pi_1(S^1) \simeq \mathbb{Z} \quad (97)$$

Proof. Let us first define the loops

$$f_n : [0, 1] \rightarrow S^1, \quad f_n(s) = (\cos 2\pi ns, \sin 2\pi ns) \quad (98)$$

We first wish to show that every loop $f : [0, 1] \rightarrow S^1$ around $x_0 = (1, 0)$ is homotopic to some f_n . Let f be some arbitrary loop. We construct the covering map $p : \mathbb{R} \rightarrow S^1$ defined $p(s) = (\cos 2\pi s, \sin 2\pi s)$. Let f be any loop in S^1 around $(1, 0)$. By the path lifting property, there exists a lift $\tilde{f} : [0, 1] \rightarrow \mathbb{R}$ starting at 0 such that the diagram commutes.

$$\begin{array}{ccc} & \mathbb{R} & \\ \tilde{f} \nearrow & \downarrow p & \\ I & \xrightarrow{f} & S^1 \end{array} \quad (99)$$

Note that $p \circ \tilde{f} = f$, so $(p \circ \tilde{f})(1) = f(1) = x_0$, which implies that $\tilde{f}(1) - \tilde{f}(0) = \tilde{f}(1) \in \mathbb{Z}$. Let us call this the *winding number* of f , denoted n_f . We claim that there is a bijective map between homotopy classes $[f]$ and winding numbers n .

1. If $f \simeq g$, then $\tilde{f}(1) = \tilde{g}(1)$, and hence $n_f = n_g$.
2. If $n_f = n_g$, then $\tilde{f}$ and $\tilde{g}$ are both paths in $\mathbb{R}$ from 0 to n . Since $\mathbb{R}$ is convex, there is a straight line homotopy between them.

Now define the straight-line path

$$\tilde{f}_n : [0, 1] \rightarrow \mathbb{R}, \quad \tilde{f}_n(s) = ns \quad (100)$$

Since $\mathbb{R}$ is convex, we know that $\tilde{f} \simeq \tilde{f}_{n_f}$ by the linear homotopy. By composing it with continuous p , we have $f = p \circ \tilde{f} \simeq p \circ \tilde{f}_{n_f} = f_{n_f}$. Now to prove the group property structure, we note that $[f_n] = [f_1]^n$. This implies that

$$n_{f*g} = n_f + n_g \quad (101)$$

Thus, the winding number gives us an isomorphism from $\pi_1(S^1) \rightarrow \mathbb{Z}$.

An application of this theorem is the proof of the fundamental theorem of algebra.

Theorem 1.21 (Fundamental Theorem of Algebra)

Every nonconstant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.

Proof. Assume on the contrary that it doesn't. We can assume the polynomial is of form $p(z) = z^n + a_1 z^{n-1} + \dots + a_n$.

1. Define the family of loops on S^1

$$f_r(s) = \frac{p(re^{2\pi is})/p(r)}{p(re^{2\pi is})/p(r)}, \quad r \geq 0 \quad (102)$$

This is indeed a loop since $F(r, s) = f_r(s)$ is continuous, and

$$f_r(0) = \frac{p(r)/p(r)}{|p(r)/p(r)|} = 1, \quad f_r(1) = \frac{p(re^{2\pi i})/p(r)}{|p(re^{2\pi i})/p(r)|} = 1 \quad (103)$$

This makes $r \mapsto F(r, s)$ a homotopy of loops relative to endpoint 1. But when $r = 0$, $f_r(s) = \frac{p(0)/p(0)}{|p(0)/p(0)|} = 1$, the constant path, which means that $[f_r] = [0]$.

2. Now we wish to show that through another homotopy, $f_r \simeq \omega_n$, where $\omega_n(s) = e^{2\pi i s}$. Fix r and consider the path homotopy with respect to the endpoints.

$$F(s, t) = \frac{p_t(re^{2\pi i s})/p_t(r)}{|p_t(re^{2\pi i s})/p_t(r)|}, \quad p_t(z) = z_n + t(a_1 z^{n-1} + \dots + a_n) \quad (104)$$

This is continuous since the map $(r, s, t) \mapsto F(r, s, t)$ is continuous, and

$$F(0, t) = \frac{p_t(r)/p_t(r)}{|p_t(r)/p_t(r)|} = 1, \quad F(1, t) = \frac{p_t(re^{2\pi i})/p_t(r)}{|p_t(re^{2\pi i})/p_t(r)|} = 1 \quad (105)$$

Now as we dial t from 1 to 0, we go from $F(s, 1) = f_r(s)$ to

$$F(s, 0) = \frac{p_0(re^{2\pi i s})/p_0(r)}{|p_0(re^{2\pi i s})/p_0(r)|} = \frac{r^n e^{2\pi i s n} / r^n}{|r^n e^{2\pi i s n} / r^n|} = e^{2\pi i s n} = \omega_n(s) \quad (106)$$

Therefore, $f_r \simeq \omega_n$, and so $[f_r]$ represents n times a generator of the infinite cyclic group $\pi_1(S^1)$. Since $f_r \simeq 0$ and $f_r \simeq \omega_n$, we arrive at a contradiction.

Theorem 1.22 (Fundamental Groups of S^n for $n \geq 2$)

When $n \geq 2$, S^n is simply connected.

$$\pi_1(S^n) \simeq \{e\} \quad (107)$$

Proof. By using the function $p : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{0\}$, given by $p(r, t) = (r \cos 2\pi t, r \sin 2\pi t)$, we can use the same logic. Another way to see it is that $\mathbb{R}^2 \setminus \{0\} \cong S^1 \times (0, \infty) \cong S^1 \times \mathbb{R}$.

Now we explore the fundamental groups of S^n minus a point. This becomes very simple once we know the stereographic projection.

Theorem 1.23 (Stereographic Projection)

We claim

$$S^n \setminus \{n\} \cong \mathbb{R}^n \quad (108)$$

Proof. We can construct a formula, but visually, we can take any $x \in S^n \setminus \{n\}$, and there is a unique line L through x and n . Now let $f(x) := L_x \cap (\mathbb{R}^{n-1} \times \{0\})$, which is bijective. It also turns out to be a homeomorphism.

Corollary 1.24 (Fundamental Groups of S^n Minus a Point)

We claim that $S^n \setminus \{p\}$ is simply connected, for all $n \in \mathbb{N}$.

$$\pi_1(S^n) \simeq \{e\} \quad (109)$$

So informally, S^n minus a point gives us $\mathbb{R}^n$, so what if we remove a point from $\mathbb{R}^n$? It turns out that by taking n points away from S^n , it is equal to taking $n - 1$ points away from $\mathbb{R}^n$.

Theorem 1.25 (Fundamental Groups of $\mathbb{R}^n$ Minus a Point)

We claim that

1. For $n = 1$, $\mathbb{R} \setminus \{0\}$ is not even path connected. For any x_0 , we have

$$\pi_1(\mathbb{R} \setminus \{0\}) \simeq \{e\} \quad (110)$$

2. For $n = 2$, $\mathbb{R}^2 \setminus \{0\}$ is path connected, but it is not simply connected.^a

$$\pi_1(\mathbb{R}^2 \setminus \{x\}) \simeq \mathbb{Z} \quad (111)$$

3. For $n \geq 3$, $\mathbb{R}^n \setminus \{x\}$ is simply connected.^b

^aWe can think of this as a hole in the middle of the plane that paths cannot unwind through.

^bWe can interpret as if the extra dimensions being added gives more space for a path to unwind.

Proof. For $n = 1$, we can see that $(-\infty, 0)$ and $(0, +\infty)$ are convex subsets, so they are simply connected. For $n = 2$, we can see in polar coordinates,

$$\mathbb{R}^2 \setminus \{0\} \simeq S^1 \times (0, +\infty) \implies \pi_1(\mathbb{R}^2 \setminus \{0\}) \simeq \pi_1(S^1) \times \pi_1((0, +\infty)) \simeq \mathbb{Z} \times \{e\} \simeq \mathbb{Z} \quad (112)$$

For $n = 3$, we can construct it from smaller simply connected spaces. Take

$$U = \mathbb{R}^n \setminus \{(0, \dots, 0, z) \mid z \geq 0\} \quad (113)$$

$$V = \mathbb{R}^n \setminus \{(0, \dots, 0, z) \mid z \leq 0\} \quad (114)$$

These are each open (since the axes that we minus out is closed) and simply connected (since they are star convex). Also, $U \cap V = (\mathbb{R}^{n-1} \setminus \{0\}) \times \mathbb{R}$, which is path connected since we assumed that $n \geq 3$. Therefore, by the theorem above, $\mathbb{R}^n \setminus \{0\}$ is simply connected.

Theorem 1.26 (Fundamental Groups of $\mathbb{RP}^n$)

We claim that.

1. For $n = 1$, $\mathbb{RP}^1 \cong S^1$ and so $\pi_1(\mathbb{RP}) \simeq \mathbb{Z}$.
2. For $n \geq 2$, $\pi_1(\mathbb{RP}^n) \simeq \mathbb{Z}_2$

Proof. For $n = 1$, recall that S^1 is homeomorphic to $S^1/\sim$ where $x \sim -x$ through the homeomorphism $z \mapsto z^2$.

For $n \geq 2$, we claim that the projection map $p : S^n \rightarrow \mathbb{RP}^n$ defined $- \pm x \rightarrow x$ is a covering map. Furthermore, we know that S^n is simply connected (since it's homeomorphic to $\mathbb{R}^n \setminus \{0\}$), and therefore there exists a bijection from $\pi(\mathbb{RP}^n, x_0)$ with $p^{-1}(x_0) = \{\pm x_0\}$. Therefore the order of the fundamental group must be 2, and it can only be $\mathbb{Z}_2$ for all x_0 .

At this point it's pretty easy to show the fundamental groups of familiar quotient spaces.

Example 1.18 (Torus)

The torus $T \cong S^1 \times S^1$ is path connected has the fundamental group

$$\pi_1(T) \simeq \pi_1(S^1) \times \pi_1(S^1) \simeq \mathbb{Z} \times \mathbb{Z} \quad (115)$$

So it is not simply connected.

Example 1.19 (Cylinder)

The cylinder is path connected and the fundamental group is

$$\pi_1(S^1 \times [0, 1]) \simeq \pi_1(S^1) \times \pi_1([0, 1]) \simeq \mathbb{Z} \times \{e\} \simeq \mathbb{Z} \quad (116)$$

Example 1.20 (Mobius Strip)

The Mobius strip is not simply connected.

Example 1.21 (Klein Bottle)

The Klein bottle is not simply connected and it is not abelian.

1.8 Retractions

Definition 1.12 (Retraction)

If $A \subset X$, a retraction $r : X \rightarrow A$ is a continuous function such that $r(a) = a$ for all $a \in A$.

Example 1.22

The following are retractions.

$$r : \mathbb{R} \rightarrow \{x\} \subset \mathbb{R}, \quad r(x) = x \quad (117)$$

This is also a retraction.

$$r : \mathbb{R} \setminus \{0\} \rightarrow S^{n-1}, \quad x \mapsto \frac{x}{\|x\|} \quad (118)$$

Lemma 1.27 (Group Homomorphism Induced By Retraction)

Given $A \subset X$, a retraction $r : X \rightarrow A$, and the canonical injection $\iota : A \rightarrow X$,

1. The induced homomorphism $r_* : \pi_1(X, a_0) \rightarrow \pi_1(A, a_0)$ is surjective.
2. The induced homomorphism $\iota_* : \pi_1(A, a_0) \rightarrow \pi_1(X, a_0)$ is injective.

Proof.

It's pretty easy to show whether a particular example is a retraction or not. However, we will mainly be interested in the existence of retractions.

Theorem 1.28 (Unit Interval to Endpoints)

There is no retraction from $D^1 = [-1, 1]$ to $S^0 = \{-1, 1\}$.

Proof. Let $X = [a, b]$. There is no retraction $f : X \rightarrow \{a, b\}$ since $\text{Im}(f)$ must be connected. But $f(a) = a, f(b) = b$. Alternatively, we can use IVT.

Theorem 1.29 (Unit Disk to Circular Boundary)

There is no retraction from $D^2 \rightarrow S^1$.^a

^aIntuitively, we can see that the mapping has to “tear” the disk somewhere!

Proof. Assume such a retraction exists. Let f_0 be any loop around any fixed point x_0 in S^1 . f_0 is homotopic to the constant function x_0 in D through the linear homotopy.

$$f_t : [0, 1] \rightarrow D^2, \quad f_t(s) = (1 - t)f_0(s) + tx_0 \quad (119)$$

Then, we can compose this with the retraction to get $r \circ f_t : [0, 1] \rightarrow S^1$. This is a homotopy since it is continuous as the composition of continuous functions, and since S^1 is invariant under r , $r \circ f_0 = f_0(s)$ and $r \circ f_1(s) = x_0$. Therefore, this is a homotopy of f_0 to x_0 in S^1 . This implies that every loop in S^1 is null-homotopic, contradicting that $\pi_1(S^1) \simeq \mathbb{Z}$.

Theorem 1.30

Let $A \subset X$, and $x_0 \in A$, and $r : X \rightarrow A$ be a retraction. Then the induced map $r_* : \pi_1(X, x_0) \rightarrow \pi_1(A, x_0)$ is surjective.

Proof. Given retraction $r : X \rightarrow A$ and the canonical injection $\iota : A \rightarrow X$, we have $r \circ \iota = \text{id}_A$ by definition. Now see how it affects the fundamental spaces.

$$\pi_1(A, x_0) \xrightarrow{\iota_*} \pi_1(X, x_0) \xrightarrow{r_*} \pi_1(A, x_0) \quad (120)$$

Therefore $r_* \circ \iota_* = \text{id}_A$. So for any loop $[f] \in \pi_1(A)$, $[f] = r_*(\iota_*([f]))$, so $[f] \in \text{Im}(r_*)$. Also ι_* is injective.

Corollary 1.31

If X is simply connected and A is not, then there is no retraction.

Extending this a bit, remember that any continuous function $f : [a, b] \rightarrow [a, b]$ must have a fixed point by IVT. Thinking $D^1 = [-1, 1]$, we claim the following. Note that a rotation of the disk D^2 has a fixed point in the center.

Theorem 1.32 (Brouwer's Fixed Point Theorem)

Any continuous function $f : D^2 \rightarrow D^2$ has a fixed point, i.e. there exists $x \in D^2$ s.t. $f(x) = x$.^a

^aThis was first proven in the 1930s based on a non-intersection theorem. But it is non-constructive, i.e. doesn't show you how to find the point.

Proof. Suppose $f : D^2 \rightarrow D^2$ has no fixed point. Define a function $g : D^2 \rightarrow D^2$ as follows. For x , take $f(x) \neq x$ and draw a ray from $f(x)$ to x (directions matter!). Where it hits the boundary is the output $g(x)$. It is much more tedious to write out in formulas, but if you do, you can see that it's just the composition of continuous functions and is so continuous. If $x \in S^1$, then $g(x) = x$, so I've just constructed a retraction from D^2 to S^1 , which is a contradiction!

So what about higher dimensions? If we believe that there is no retraction $D^n \rightarrow S^{n-1}$, by the same method we can prove for $D^n \rightarrow D^n$. The problem is that for $n \geq 2$ S^{n-1} is simply connected and so $\pi_1(S^{n-1}) = \{e\}$,

and same holds for D^n . So we can't prove the non-retraction using fundamental groups π_1 . This is where we need higher order homotopy groups.

1.9 Exercises

Exercise 1.1 (Hatcher 1.3)

For a path-connected space X , show that $\pi_1(X)$ is abelian iff all basepoint-change homomorphisms β_h depend only on the endpoints of path h .

Solution. We prove bidirectionally.

1. ($\rightarrow$). Let $\pi_1(X)$ be abelian, with $x_0, x_1 \in X$. Let $h_1, h_2 : [0, 1] \rightarrow X$ be paths from x_0 to x_1 . Then,

$$\beta_{h_1}([f]) = [h_1 * f * \bar{h}_1] \quad (121)$$

$$= [h_1 * f * \bar{h}_2 * h_2 * \bar{h}_1] \quad (122)$$

$$= [h_1 * f * \bar{h}_2] * [h_2 * \bar{h}_1] \quad (123)$$

$$= [h_2 * \bar{h}_1] * [h_1 * f * \bar{h}_2] \quad (124)$$

$$= [h_2 * \bar{h}_1 * h_1 * f * \bar{h}_2] \quad (125)$$

$$= [h_2 * f * \bar{h}_2] \quad (126)$$

$$= \beta_{h_2}([f]) \quad (127)$$

2. ($\leftarrow$). The idea is to use two paths with the same endpoints built from loops f and g . We have

$$\beta_f([f]) = \beta_g([f]) \implies [f * f * \bar{f}] = [g * f * \bar{g}] \quad (128)$$

$$\implies [f] = [g * f * \bar{g}] \quad (129)$$

$$\implies [f] * [g] = [g * f * \bar{g}] * [g] \quad (130)$$

$$\implies [f] * [g] = [g * f * \bar{g} * g] = [g] * [f] \quad (131)$$

Exercise 1.2 (Hatcher 1.5)

Show that for a space X , the following three conditions are equivalent:

- (a) Every map $S^1 \rightarrow X$ is homotopic to a constant map, with image a point.
- (b) Every map $S^1 \rightarrow X$ extends to a map $D^2 \rightarrow X$.
- (c) $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \rightarrow X$ are homotopic. [In this problem, "homotopic" means "homotopic without regard to basepoints".]

Solution. We prove the following.

1. (a) $\implies$ (b). Let $f : S^1 \rightarrow X$ be continuous. By hypothesis, there exists a homotopy $F : S^1 \times [0, 1] \rightarrow X$ such that $F(s, 0) = x_0$ and $F(s, 1) = f(s)$. We can use polar coordinates to define the change of basis

$$q : S^1 \times [0, 1] \rightarrow D^2, \quad q(\theta, r) = (r \cos \theta, r \sin \theta) \quad (132)$$

where we parameterize S^1 with the angle θ . Since q is a quotient map, this induces a new continuous map $\bar{F} : D^2 \rightarrow X$ such that the diagram commutes.

$$\begin{array}{ccc}
S^1 \times [0, 1] & \xrightarrow{F} & X \\
\downarrow q & \nearrow \bar{F} & \\
D^2 & &
\end{array}$$

To check that $\bar{F}$ is indeed an extension of f , let $z \in S^1$. Since $q(z, 1) = z$, we have

$$\bar{F}(z) = \bar{F}(q(z, 1)) = F(z, 1) = f(z). \quad (133)$$

2. (b) $\implies$ (c). Let $f : [0, 1] \rightarrow X$ be any loop with basepoint at $x_0 \in X$. We apply the quotient map $q : [0, 1] \rightarrow S^1, q(s) = (\cos 2\pi s, \sin 2\pi s)$ and get the induced map $\bar{f} : S^1 \rightarrow X$ such that $f = \bar{f} \circ q$. Note that this means $\bar{f}(1, 0) = x_0$.

$$\begin{array}{ccc}
[0, 1] & \xrightarrow{f} & X \\
\downarrow q & \nearrow \bar{f} & \\
S^1 & &
\end{array}$$

By hypothesis, we can extend $\bar{f}$ to a continuous map $\bar{f}' : D^2 \rightarrow X$. Define $q' : [0, 1] \rightarrow D^2$ with $q'(s) = q(s)$. Since D^2 is convex, we know $q' \simeq c_{(1,0)}$ (where $c_{(1,0)}$ is the constant map to $(1, 0)$) defined by the linear homotopy $F(s, t) = (1 - t)q'(s) + t(1, 0)$ relative to the endpoints. This satisfies $F(0, t) = F(1, t) = (1, 0)$. Therefore, $\bar{f}' \circ F$ is a homotopy (relative to endpoints) from $f = \bar{f}' \circ q'$ to $\bar{f}' \circ c_{(1,0)} = x_0$.

3. (c) $\implies$ (a). Let $f : S^1 \rightarrow X$ be any continuous function, $q : [0, 1] \rightarrow S^1$ be defined $q(s) = e^{2\pi i s}$, and $g = f \circ q : [0, 1] \rightarrow X$ be the loop around $g(0) = g(1) = f(1) = x_0$. By hypothesis, we know that $g \simeq c_{x_0}$, i.e. there exists a homotopy $F : [0, 1] \times [0, 1] \rightarrow X$ such that $F(s, 0) = g(s)$, $F(s, 1) = x_0$, and $F(0, t) = F(1, t) = x_0$. We again use the induced map from the quotient that agrees with the commutative diagram below.

$$\begin{array}{ccc}
[0, 1] \times [0, 1] & \xrightarrow{F} & X \\
\downarrow q \times \text{id} & \nearrow \bar{F} & \\
S^1 \times [0, 1] & &
\end{array}$$

where $\bar{F}(q(s), t) = F(s, t)$. This is indeed a homotopy from f to x_0 since $\bar{F}(s, 0) = f(s)$ and $\bar{F}(s, 1) = x_0$.

For the forward implication, since X is simply connected, so it satisfies (c) by definition. Therefore (a) is also satisfied. Since X is path connected (since it is simply connected), any two constant maps are homotopic using a path between them. Therefore, from (a), given two functions f, g , they are each homotopic to constant map which are homotopic to each other: $f \simeq c_{x_0} \simeq c_{x_1} \simeq g$.

For the backward implication, the hypothesis immediately implies (a) since it is a special case, which then gives us (c). To prove path connectedness, note that from the hypothesis, we can take two arbitrary constant maps c_{x_0} and c_{x_1} , which are homotopic, and the homotopy is precisely the path that connects them. This establishes path connectedness, and so X is simply connected.

Exercise 1.3 (Hatcher 1.18)

Using Lemma 1.15, show that if a space X is obtained from a path-connected subspace A by attaching a cell e^n with $n \geq 2$, then the inclusion $A \hookrightarrow X$ induces a surjection on π_1 . Apply this to show:

- The wedge sum $S^1 \vee S^2$ has fundamental group $\mathbb{Z}$.
- For a path-connected CW complex X , the inclusion map $X^1 \hookrightarrow X$ of its 1-skeleton induces a surjection

$$\pi_1(X^1) \rightarrow \pi_1(X).$$

[For the case that X has infinitely many cells, see Proposition A.1 in the Appendix.]

Solution. Let us attach at A the n -cell e^n by taking some equivalence relation using a function $\varphi : \partial e^n \rightarrow A$ and defining

$$X = \frac{e^n \sqcup A}{(x \sim \varphi(x), x \in \partial e^n)} \quad (134)$$

which is path connected as the union of two path connected sets. The function that constructs this set is the quotient map $q : e^n \sqcup A \rightarrow X$. We would like to represent X as a union of two path-connected open sets U and V , but A is not guaranteed to be open.^a

Therefore, we thicken A into the cell by taking some small $\epsilon > 0$ and defining

$$U = q((A \sqcup \{x \in e^n : \|x\| > 1 - \epsilon\}) / \sim) \quad (135)$$

$$V = q(\{x \in e^n : \|x\| < 1 - \frac{\epsilon}{2}\}) \quad (136)$$

Colloquially, U is “ A with the collar” and V is a slightly smaller open subset of the n -cell. We can see that $X = U \cup V$ with U, V open and path connected (since continuous images of path connected spaces are path connected). The intersection is

$$U \cap V = q(\{x \in e^n : 1 - \epsilon < \|x\| < 1 - \frac{\epsilon}{2}\}) \quad (137)$$

is also path connected. Since A and X are both path connected, their fundamental groups don't depend on the basepoint, and we can investigate the induced homomorphism of the inclusion map $\iota_* : \pi_1(A) \rightarrow \pi_1(X)$. Choose a basepoint $x_0 \in U \cap V$ and take any loop $f : [0, 1] \rightarrow X$ about basepoint x_0 .

1. If the image of f is in U , then since U deformation retracts onto A , the loop f is homotopic in U to a loop in A , and we have found a preimage of this path.
2. If the image of f is not completely in U , we can invoke Lemma 1.15 and decompose f into the sequence of loops

$$f \simeq (f_1 * \bar{g}_1) * (g_1 * f_2 * \bar{g}_2) * \dots * (g_{m-1} * f_m) \quad (138)$$

where each term (the loop in one set of parentheses) is completely in U or V . If a term lies in V , then it is null-homotopic in V , since V is contractible. If a term lies in U , then by the same logic above, it is homotopic to a loop in A . Therefore, by remove the terms in V , we can find a loop completely contained in A , i.e. the preimage.

Now to apply this result:

1. Setting $A = S^1$ and $e^2 = D^2$. This inclusion induces a surjection from $\pi_1(S^1) \simeq \mathbb{Z}$ to $\pi_1(S^1 \vee S^2)$. The attaching map $\varphi : \partial D^2 = S^1 \rightarrow S^1$ is constant, so the attaching loop represents the trivial element of $\pi_1(S^1)$. Hence $\pi_1(S^1 \vee S^2) = \pi_1(S^1) \simeq \mathbb{Z}$.
2. Since X is built skeleton by skeleton, we have

$$X^1 \subset X^2 \subset X^3 \subset \dots \subset X. \quad (139)$$

The space X^2 is obtained from X^1 by attaching 2-cells. X^1 may not be path connected, so we just do this construction to all the path connected components of X^1 . So by the previous result, the inclusion induces a surjection

$$\pi_1(X^1) \rightarrow \pi_1(X^2). \quad (140)$$

More generally, X^{k+1} is obtained from X^k by attaching $(k+1)$ -cells. Since $k+1 \geq 2$, we also get surjections $\pi_1(X^k) \rightarrow \pi_1(X^{k+1})$. Applying this inductively, we obtain surjections

$$\pi_1(X^1) \rightarrow \pi_1(X^2) \rightarrow \pi_1(X^3) \rightarrow \dots, \quad (141)$$

Since the composition of surjections is a surjection, the inclusion $\iota : X^1 \rightarrow X$ induces a surjection $\pi_1(X^1) \rightarrow \pi_1(X)$.

^aSince $q^{-1}(A) = A \sqcup \partial e^n$, where A is open in the disjoint union, but ∂e^n is not open in e^n .

Exercise 1.4 (Hatcher 1.20)

Suppose $f_t : X \rightarrow X$ is a homotopy such that f_0 and f_1 are each the identity map. Use Lemma 1.19 to show that for any $x_0 \in X$, the loop $f_t(x_0)$ represents an element of the center of $\pi_1(X, x_0)$. [One can interpret the result as saying that a loop represents an element of the center of $\pi_1(X)$ if it extends to a loop of maps $X \rightarrow X$.]

Solution. Let $h(t) = f_t(x_0)$. Then, $h : [0, 1] \rightarrow X$ is a loop around basepoint x_0 , and

$$\beta_h : \pi_1(X, x_0) \rightarrow \pi_1(X, x_0), \quad \beta_h([f]) = [h * f * \bar{h}] \quad (142)$$

is the induced homomorphism. By Lemma 1.19, β_h must satisfy

$$f_{0*} = \beta_h \circ f_{1*} \implies \text{id} = \beta_h \circ \text{id} \quad (143)$$

Therefore, for any loop $f : [0, 1] \rightarrow X$ about basepoint x_0 ,

$$\beta_h = \text{id} \implies \beta_h([f]) = [h * f * \bar{h}] = [f] \quad (144)$$

$$\implies [h] * [f] * [\bar{h}] = [f] \quad (145)$$

$$\implies [h] * [f] = [f] * [h] \quad (146)$$

Therefore, the loop h commutes with every element in $\pi_1(X, x_0)$, which is by definition means it is in the center of $\pi_1(X, x_0)$.

Exercise 1.5 (Munkres 51.1)

Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

Solution. From the assumptions there exists homotopies $H : X \times I \rightarrow Y$ and $K : Y \times I \rightarrow Z$. We define the composition homotopy as

$$G : X \times I \rightarrow Z, \quad G(x, t) := K(H(x, t), t) \quad (147)$$

Note that since $(x, t) \mapsto H(x, t)$ is continuous, by the product topology the map $(x, t) \mapsto (H(x, t), t)$ is continuous, and it composed with continuous K makes it also continuous. Finally, we have

$$G(x, 0) = K(H(x, 0), 0) = K(h(x), 0) = k(h(x)) = (k \circ h)(x) \quad (148)$$

$$G(x, 1) = K(H(x, 1), 1) = K(h'(x), 1) = k'(h'(x)) = (k' \circ h')(x) \quad (149)$$

Exercise 1.6 (Munkres 51.2)

Given spaces X and Y , let $[X, Y]$ denote the set of homotopy classes of maps of X into Y .

(a) Let $I = [0, 1]$. Show that for any X , the set $[X, I]$ has a single element.

(b) Show that if Y is path connected, the set $[I, Y]$ has a single element.

Solution. Listed.

1. Take continuous $f, g : X \rightarrow I$ and define the homotopy $F : X \times I \rightarrow I$ between them as the convex combination

$$F(x, t) = (1 - t)f(x) + tg(x) \quad (150)$$

We have $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$, and the products/sums of continuous maps are

continuous. Finally, we can see that $F(x, t) \in I$ because given any x ,

$$f(x) \leq g(x) \implies f(x) \leq F(x, t) = (1-t)f(x) + tg(x) \leq g(x) \quad (151)$$

$$f(x) \geq g(x) \implies g(x) \leq F(x, t) = (1-t)f(x) + tg(x) \leq f(x) \quad (152)$$

for all $t \in [0, 1]$, and since $[0, 1] \subset \mathbb{R}$ is convex, $F(x, t) \in [0, 1]$ for all t and x .

2. Given any continuous function $f : I \rightarrow Y$, this is by definition a path. We wish to show that this is homotopic to a constant map, say the value at $f(0) \in Y$. We construct the homotopy

$$F(x, t) : I \times I \rightarrow Y, \quad F(x, t) = f((1-t)x) \quad (153)$$

This satisfies $F(x, 0) = f(x)$ and $F(x, 1) = f(0)$. This is also continuous since it is the composition of continuous $(x, t) \mapsto (1-t)x$ and continuous f . Therefore, given two functions $f, g : I \rightarrow Y$, we know that $f \simeq f(0)$ and $g \simeq g(0)$. Now since homotopy is an equivalence relation, it suffices to show that $f(0) \simeq g(0)$, i.e. any two constant functions are homotopic. Since Y is path connected, there exists a path $h : I \rightarrow Y$ such that $h(0) = f(0), h(1) = g(0)$. Now we define the homotopy

$$H : I \times I \rightarrow Y, \quad H(x, t) = h(t) \quad (154)$$

which is certainly continuous since path h is continuous by definition, and $h(0) = f(0), h(1) = g(0)$.

Exercise 1.7 (Munkres 51.3)

A space X is said to be *contractible* if the identity map $i_X : X \rightarrow X$ is nullhomotopic.

- Show that I and $\mathbb{R}$ are contractible.
- Show that a contractible space is path connected.
- Show that if Y is contractible, then for any X , the set $[X, Y]$ has a single element.
- Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Solution. Listed.

- For I , we define the homotopy $F(x, t) = (1-t)x$, which is continuous (since it is a scalar multiple of the continuous identity map) and satisfies $F(x, 0) = x = i_X(x)$, and $F(x, 1) = 0$ which is a constant map. For $\mathbb{R}$ we can use the same homotopy and the same logic follows.
- Given that X is nullhomotopic let $z \in X$ be the value of the constant map that is homotopic to the identity $i : X \rightarrow X$, with homotopy $F : X \times I \rightarrow X$ where $F(x, 0) = x$ and $F(x, 1) = z$. Then we claim that for any $x \in X$, x is path connected to z since by fixing x , the function $\gamma : I \rightarrow X$ defined $\gamma(t) = F(x, t)$ is a path: it is continuous, $\gamma(0) = F(x, 0) = x$, and $\gamma(1) = F(x, 1) = z$. Therefore we have found such a path, and since path connectedness is an equivalence relation, for any $x, y \in X$, $x \sim z \sim y$, making X path connected.
- Since Y is contractible, let us construct the homotopy of the constant map to the identity map $F : Y \times I \rightarrow Y$ satisfying $F(y, 0) = z$ for some $z \in Y$ and $F(y, 1) = y$. Now given continuous $g : X \rightarrow Y$, we wish to show that it is homotopic to the constant map with value z . Consider the homotopy $G : X \times I \rightarrow Y$ defined

$$G(x, t) = F(g(x), t) \quad (155)$$

This is clearly continuous as the composition of continuous maps. Furthermore, $G(x, 0) = F(g(x), 0) = z$ and $G(x, 1) = F(g(x), 1) = g(x)$, making this indeed a homotopy. Since homotopy is an equivalence relation for any continuous $f, g : X \rightarrow Y$, $f \simeq z \simeq g$, and so there is only one equivalence class.

- We wish to show that given any two functions $f, g : X \rightarrow Y$, $f \simeq g$. Since X is contractible, there exists a homotopy $F : X \times I \rightarrow X$ where $F(x, 0) = z$ for some $z \in X$ and $F(x, 1) = x$. We can take f, g and see that $(f \circ F), (g \circ F) : X \times I \rightarrow Y$ is a homotopy, as the composition of

continuous maps satisfying

$$(f \circ F)(x, 0) = f(F(x, 0)) = f(z) \quad (156)$$

$$(f \circ F)(x, 1) = f(F(x, 1)) = f(x) \quad (157)$$

$$(g \circ F)(x, 0) = g(F(x, 0)) = g(z) \quad (158)$$

$$(g \circ F)(x, 1) = g(F(x, 1)) = g(x) \quad (159)$$

Therefore, $f(z) \sim f$ and $g(z) \sim g$. We know that since Y is path connected, $f(z) \sim g(z)$ as given in my argument in Munkres 51.2.b, and so $f \sim f(z) \sim g(z) \sim g$, and every continuous function is a part of the same equivalence class.

2 Other

We can also expect that since homotopy is clearly a topological property, it is preserved under continuous maps. We state this result formally in the following lemma.

Lemma 2.1

If $F_0, F_1 : X \rightarrow Y$ and $G_0, G_1 : Y \rightarrow Z$ are continuous maps such that $F_0 \simeq F_1$ and $G_0 \simeq G_1$, then

$$G_0 \circ F_0 \simeq G_1 \circ F_1$$

Similarly, if $f_0, f_1 : I \rightarrow X$ are path homotopic, and $F : X \rightarrow Y$ is a continuous map, then

$$F \circ f_0 \sim F \circ f_1$$

Thus, if $F : X \rightarrow Y$ is a continuous maps, for each $q \in X$, we can construct a well-defined map

$$F_* : \pi_1(X, q) \rightarrow \pi_1(Y, F(q))$$

by setting

$$F_*([f]) \equiv [F \circ f]$$

Lemma 2.2

If $F : X \rightarrow Y$ is a continuous map, then the induced map

$$F_* : \pi_1(X, q) \rightarrow \pi_1(Y, F(q))$$

is a group homomorphism. That is, F_* preserves multiplicative structure of the loops.

Theorem 2.3 (Properties of the Induced Homomorphism)

1. Let $F : X \rightarrow Y, G : Y \rightarrow Z$ be continuous maps. Then for any $q \in X$,

$$(G \circ F)_* = G_* \circ F_* : \pi_1(X, q) \rightarrow \pi_1(Z, G(F(q)))$$

2. For any space X and any $q \in X$, the homomorphism induced by the identity map $id_X : X \rightarrow X$ is the identity map

$$id : \pi_1(X, q) \rightarrow \pi_1(X, q)$$

3. If $F : X \rightarrow Y$ is a homeomorphism, then

$$F_* : \pi_1(X, q) \rightarrow \pi_1(Y, F(q))$$

is an isomorphism. That is, homeomorphic spaces have isomorphic fundamental groups.

Example 2.1

The fundamental group of $S^1 \subset \mathbb{C}$ based at 0 is the infinite cyclic group generated by the path class of the loop

$$\alpha : I \longrightarrow S^1, \alpha(s) \equiv e^{2\pi is}$$

Theorem 2.4

If $F : X \longrightarrow Y$ is a homotopy equivalence, then for each $p \in X$,

$$F_* : \pi_1(X, p) \longrightarrow \pi_1(Y, F(p))$$

is an isomorphism.

The following theorem will be revisited when studying manifolds.

Theorem 2.5

The fundamental group of any topological manifold is countable.

2.1 Homotopy Equivalence

Definition 2.1

Given two topological spaces X and Y , a homotopy equivalence between X and Y is a pair of continuous maps $f : X \longrightarrow Y$ and $g : Y \longrightarrow X$ such that

$$g \circ f \simeq id_X \text{ and } f \circ g \simeq id_Y$$

The equivalence classes under $\simeq$ are called *homotopy types*. If such a pair f, g exists, X and Y are said to be *homotopy equivalent*, or of the same homotopy type.

Definition 2.2

Spaces that are homotopy equivalent to a point are called *contractible*. That is, X is contractible if and only if

$$X \simeq \{x_0\}$$

Visually, two spaces are homotopy equivalent if they can be transformed into one another by bending, shrinking, and expanding operations.

Example 2.2

A solid disk is homotopy equivalent to a single point, since one can deform the disk along radial lines to a point.

Example 2.3

A mobius strip is homotopy equivalent to a closed (untwisted) strip.

Notice from the visualization of homotopy equivalence the following theorem.

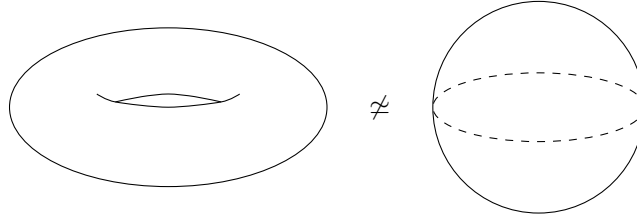
Theorem 2.6

X, Y homeomorphic $\implies X, Y$ homotopy equivalent. However, the converse is not true.

Proof. Just set $f = f$ and $g = f^{-1}$.

Example 2.4

A torus is not homotopy equivalent to Y , which also implies that they are not homeomorphic either.



Furthermore, like homeomorphisms, homotopy equivalence is a relation on the set of all topological spaces.

1. Identity. Just set $f, g = id_X$
2. Symmetricity. Given $X \simeq Y$ with $f : X \longrightarrow Y, g : Y \longrightarrow X$, we set $f' \equiv g$ and $g' \equiv f$ and use these functions f', g' to find out that $Y \simeq X$.
3. Transitivity. Let us have $X \simeq Y$ with functions f_1, g_1 and $Y \simeq Z$ with functions f_2, g_2 . Then, we define new functions

$$f_3 \equiv f_2 \circ f_1 : X \longrightarrow Z, g_3 \equiv g_1 \circ g_2 : Z \longrightarrow X \quad (160)$$

which follows to $f_3 \circ g_3 = id_Z$ and $g_3 \circ f_3 = id_X$.

Theorem 2.7

$\mathbb{R}^n$ is homotopically equivalent to a point $\{0\}$.

Proof. We claim that the continuous maps (canonical injection and projection)

$$id_{\mathbb{R}^n} : \{0\} \longrightarrow \mathbb{R}^n, p_0 : \mathbb{R}^n \longrightarrow \{0\} \quad (161)$$

have the property that

$$id_{\mathbb{R}^n} \circ p_0 \simeq id_{\mathbb{R}^n}, p_0 \circ id_{\mathbb{R}^n} \simeq id_{\{0\}} \quad (162)$$

The right-hand homotopy is trivial since $id_{\mathbb{R}^n} \circ p_0 = id_{\mathbb{R}^n}$, and as for the left-hand homotopy, we can explicitly define it as

$$H : [0, 1] \times \mathbb{R}^n \longrightarrow \mathbb{R}^n \quad (163)$$

with

$$H(t, x) \equiv (t)(id_{\mathbb{R}^n} \circ p_0)(x) + (1 - t)id_{\mathbb{R}^n}(x) = (1 - t)id_{\mathbb{R}^n}(x) \quad (164)$$

Example 2.5

$S^1 \simeq \mathbb{R}^2 \setminus \{0\}$, and more generally, $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$. We can see this with the canonical injection and projections

$$id_{\mathbb{R}^2} : S^1 \longrightarrow \mathbb{R}^2 \setminus \{0\}, \pi_{S^1} : \mathbb{R}^2 \setminus \{0\} \longrightarrow S^1 \quad (165)$$

and find that

$$id_{\mathbb{R}^2} \circ \pi_{S^1} \simeq id_{\mathbb{R}^2}, \pi_{S^1} \circ id_{\mathbb{R}^2} \simeq id_{S^1} \quad (166)$$

where the right-hand homotopy is trivial, and the left hand homotopy is defined explicitly as

$$H(x, t) \equiv t(id_{\mathbb{R}^2} \circ \pi_{S^1})(x) + (1 - t)(id_{\mathbb{R}^2})(x) \quad (167)$$

Example 2.6

A map $f : S^1 \longrightarrow X$ is null homotopic precisely when it can be continuously extended to a map

$$\tilde{f} : D^2 \longrightarrow X \quad (168)$$

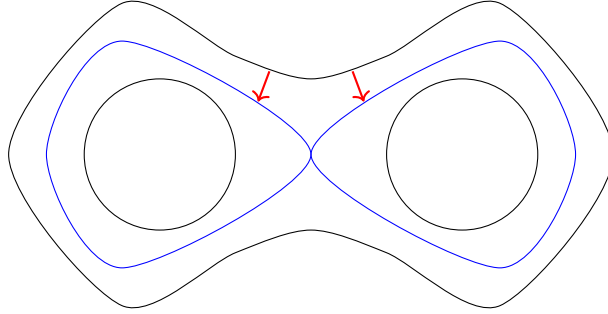
that agrees with f on the boundary $\partial D^2 = S^1$. Visually, the existence of $\tilde{f}$ allows us to continuously deform the image of f in $S^1 \oplus X$ to a level curve $f(x) = c$ existing in $S^1 \oplus X$.

Theorem 2.8

A space X is contractible if and only if the identity map from X to itself, which is always a homotopy equivalence, is null homotopic.

Example 2.7

Let Y be the following gray subset of the plane, and let X be the figure-8 shape.



Then $Y \simeq X$, where the corresponding functions are

$$F : X \longrightarrow Y, \text{ the canonical inclusion} \quad (169)$$

$$F : Y \longrightarrow X, \text{ the projection onto } X \quad (170)$$

Then, $G \circ F = id$ and $F \circ G$ is homotopic to the identity, with homotopy defined

$$H(x, t) \equiv t(F \circ G)(x) + (1 - t)(id_Y)(x) \quad (171)$$

which can be visualized by $H(x, s)$ being the point you get from x by moving a fraction s along the red arrow towards X .

3 Homeomorphism Groups

Definition 3.1 (Homeomorphism Group)

The **homeomorphism group** of a topological space X is the group consisting of all homeomorphisms from X to X , with function composition as the group operation.